

Eigenvalues and eigenvectors continued...

Recall:

Definition 27. Let A be an $n \times n$ matrix. A scalar λ is called an eigenvalue of matrix A if there is a vector $\mathbf{x} \neq \mathbf{0}$ such that $A\mathbf{x} = \lambda\mathbf{x}$, this non-zero vector $\mathbf{x}$ is called an eigenvector corresponding to the eigenvalue λ .

The following theorem shall help us in finding eigenvalues of a matrix.

Theorem 7. Let A be an $n \times n$ matrix, then λ is an eigenvalue of A if and only if $\det(\lambda I_n - A) = 0$.

Proof. Notice that we have $A\mathbf{x} = \lambda\mathbf{x} \iff \lambda\mathbf{x} - A\mathbf{x} = \mathbf{0} \iff (\lambda I_n - A)\mathbf{x} = \mathbf{0}$, the linear system $(\lambda I_n - A)\mathbf{x} = \mathbf{0}$ has a non-zero solution $\iff$ matrix $\lambda I_n - A$ is singular $\iff$ matrix $\lambda I_n - A$ is not invertible $\iff \det(\lambda I_n - A) = 0$. $\square$

Example Find the eigenvalues of matrix $A = \begin{pmatrix} 1 & 2 & 0 \\ -1 & -1 & 1 \\ 0 & 1 & 1 \end{pmatrix}$ and for each eigenvalue, find any corresponding eigenvector.

We have that $\lambda I_3 - A = \begin{pmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{pmatrix} - \begin{pmatrix} 1 & 2 & 0 \\ -1 & -1 & 1 \\ 0 & 1 & 1 \end{pmatrix} = \begin{pmatrix} \lambda - 1 & -2 & 0 \\ 1 & \lambda + 1 & -1 \\ 0 & -1 & \lambda - 1 \end{pmatrix}$ and

$\det(\lambda I_3 - A) = \begin{vmatrix} \lambda - 1 & -2 & 0 \\ 1 & \lambda + 1 & -1 \\ 0 & -1 & \lambda - 1 \end{vmatrix}$, expanding using the first column we get

$\det(\lambda I_3 - A) = (\lambda - 1) \begin{vmatrix} \lambda + 1 & -1 \\ -1 & \lambda - 1 \end{vmatrix} - 1 \cdot \begin{vmatrix} -2 & 0 \\ -1 & \lambda - 1 \end{vmatrix} + 0 \cdot \begin{vmatrix} -2 & 0 \\ \lambda + 1 & -1 \end{vmatrix}$ and so

$\det(\lambda I_3 - A) = (\lambda - 1)(\lambda^2 - 2) + 2(\lambda - 1) = (\lambda - 1)(\lambda^2 - 2 + 2) = (\lambda - 1)\lambda^2$,

hence $\det(\lambda I_3 - A) = 0 \iff \lambda = 0$ or $\lambda = 1$.

To find eigenvector associated to eigenvalue $\lambda = 0$, we need to solve linear system $A\mathbf{x} = 0\mathbf{x} = \mathbf{0}$, equivalently, $(0 \cdot I_3 - A)\mathbf{x} = \mathbf{0}$

$$\begin{aligned} \left(\begin{array}{ccc|c} -1 & -2 & 0 & 0 \\ 1 & 1 & -1 & 0 \\ 0 & -1 & -1 & 0 \end{array} \right) &\xrightarrow{R_1 \leftrightarrow R_2} \left(\begin{array}{ccc|c} 1 & 1 & -1 & 0 \\ -1 & -2 & 0 & 0 \\ 0 & -1 & -1 & 0 \end{array} \right) \\ &\xrightarrow{R_2 \rightarrow R_2 + R_1} \left(\begin{array}{ccc|c} 1 & 1 & -1 & 0 \\ 0 & -1 & -1 & 0 \\ 0 & -1 & -1 & 0 \end{array} \right) \\ &\xrightarrow{R_3 \rightarrow R_3 - R_2} \left(\begin{array}{ccc|c} 1 & 1 & -1 & 0 \\ 0 & -1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right) \end{aligned}$$

Notice that x_3 is a free variable, let $x_3 = \mu$ where $\mu \in \mathbb{R}$, we also have $-x_2 - x_3 = 0$, $x_1 + x_2 - x_3 = 0$ and so $x_2 = -\mu$ and $x_1 = 2\mu$. In vector form, the solution is $\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \mu \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$ (where $\mu \in \mathbb{R}$) and since the question asks for any eigenvector, we can simply take non-zero vector $\mathbf{v} = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$ in the solution set as our eigenvector.

We now find eigenvector associated to eigenvalue $\lambda = 1$, which we can do by solving linear system $(1 \cdot I_3 - A)\mathbf{x} = \mathbf{0}$

$$\begin{pmatrix} 0 & -2 & 0 & | & 0 \\ 1 & 2 & -1 & | & 0 \\ 0 & -1 & 0 & | & 0 \end{pmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{pmatrix} 1 & 2 & -1 & | & 0 \\ 0 & -2 & 0 & | & 0 \\ 0 & -1 & 0 & | & 0 \end{pmatrix} \xrightarrow{R_3 \rightarrow R_3 - \frac{1}{2}R_2} \begin{pmatrix} 1 & 2 & -1 & | & 0 \\ 0 & -2 & 0 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{pmatrix}$$

We have that x_3 is a free variable, let $x_3 = \mu$ where $\mu \in \mathbb{R}$, we also have $-2x_2 = 0$, $x_1 + 2x_2 - x_3 = 0$ and so $x_2 = 0$ and $x_1 = \mu$. In vector form, the solution is $\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \mu \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$ (where $\mu \in \mathbb{R}$) and we can take non-zero vector $\mathbf{v} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$ in the solution set as our eigenvector.

More properties of determinants and Cramer's rule

Recall that given an $n \times n$ elementary matrix E and an $n \times n$ matrix A , we have that $\det(EA) = \det(E)\det(A)$, we shall use this result to prove a more general result below.

Theorem 8. *Let A and B be $n \times n$ matrices, then $\det(AB) = \det(A)\det(B)$.*

Proof. We consider two exhaustive cases:

Case (1): If A is singular, then AB is also singular (for if not, that is, if AB is non-singular we would have both A and B non-singular) we then have

$$\det(A)\det(B) = 0 \cdot \det(B) = 0 = \det(AB)$$

Case (2): If A is non-singular, then by the invertibility matrix theorem it can be written as a product of elementary matrices, that is, there exists elementary matrices $E_1, E_2, \dots, E_k$ such that $A = E_k \cdots E_2 E_1$ and so we have $\det(AB) = \det(E_k \cdots E_2 E_1 B) = \det(E_k) \cdots \det(E_2) \det(E_1) \det(B) = \det(E_k \cdots E_2 E_1) \det(B) = \det(A) \det(B)$. $\square$

Corollary 3. *Let A be an $n \times n$ matrix such that $\det(A) \neq 0$, then*

$$\det(A^{-1}) = \frac{1}{\det(A)}$$

Proof. Let A be an $n \times n$ matrix such that $\det(A) \neq 0$, then A is invertible and from $A^{-1}A = I_n$ we get $\det(A^{-1})\det(A) = \det(I_n) = 1$ and so $\det(A^{-1}) = \frac{1}{\det(A)}$. $\square$

Let A be an $n \times n$ matrix and $\mathbf{b}$ a column vector with n -entries, we shall denote by $A_k[\mathbf{b}]$ (where $1 \leq k \leq n$) the matrix obtained by replacing the k -th column of A by $\mathbf{b}$.

Example If $A = \begin{pmatrix} 1 & 0 & 4 \\ 2 & 6 & 9 \\ \pi & 7 & 3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 1 \\ 7 \\ 4 \end{pmatrix}$ then $A_2[\mathbf{b}] = \begin{pmatrix} 1 & 1 & 4 \\ 2 & 7 & 9 \\ \pi & 4 & 3 \end{pmatrix}$

Theorem 9. (Cramer's rule) Let A be an invertible $n \times n$ matrix and $\mathbf{b} \in \mathbb{R}^n$, then the linear system $A\mathbf{x} = \mathbf{b}$ has a unique solution and is given by

$$x_k = \frac{\det(A_k[\mathbf{b}])}{\det(A)}, \text{ for } k=1,2,\dots, n.$$

...We shall continue in the next lecture...